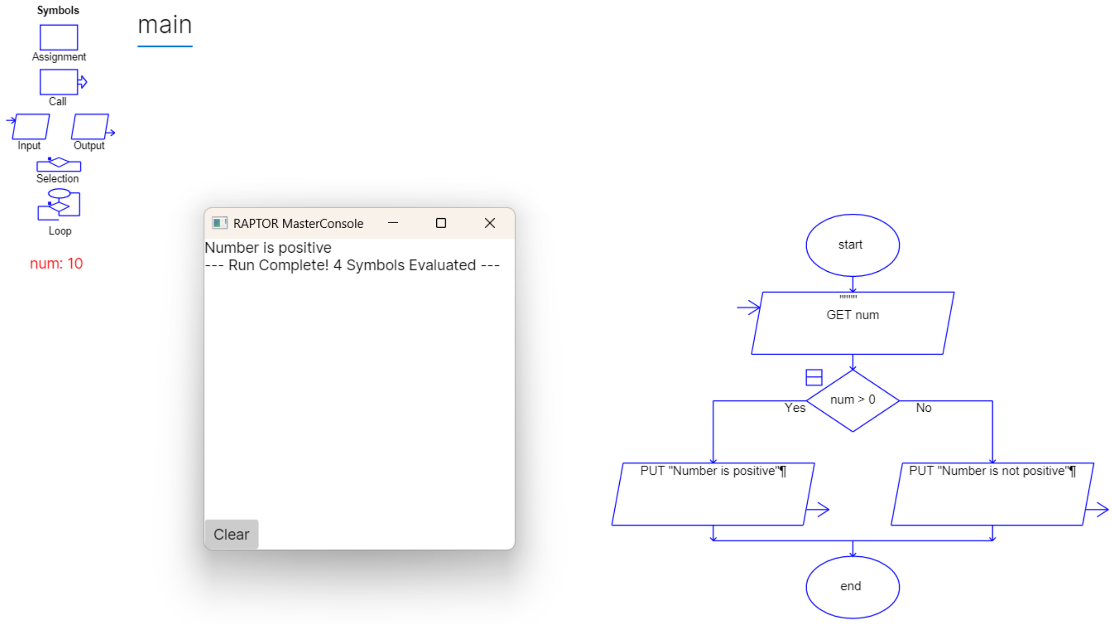
CDAC Mumbai
Lab Assignment: Flowchart and Java Programming

1. Check Positive Number:

- **Task:** Create a flowchart to check whether a number is positive.
- **Next Step:** Write a Java program that checks if a predefined number is positive using an if-else statement and prints the appropriate message.

Flowchart:



Java Code:

```
public class PositiveNumber {
	public static void main(String[] args) {
		int num = 18;
		if (num > 0) {
			System.out.println("Number is Positive");
		}
		else {
			System.out.println("Number is not positive");
		}
	}
}
```

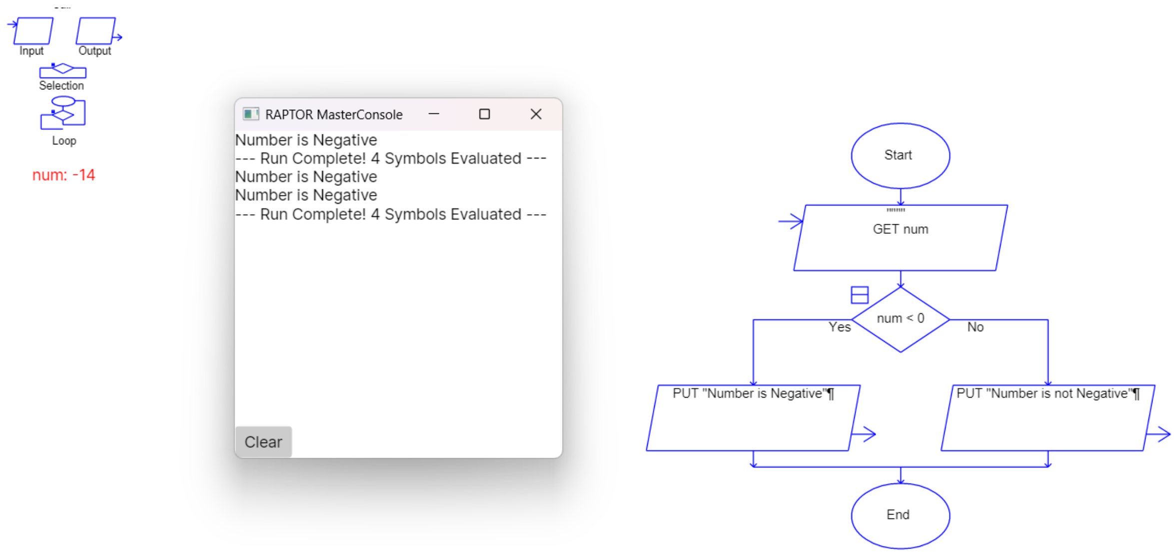
Output:

```
PS F:\Java Code> javac PositiveNumber.java
PS F:\Java Code> java PositiveNumber
Number is Positive
```

2. Check Negative Number:

- **Task:** Create a flowchart to check whether a number is negative.
- **Next Step:** Write a Java program that checks if a predefined number is negative using an if-else statement and displays the result.

Flowchart:



Java Code:

```
public class NegativeNumber {
        public static void main(String[] args) {
            int num = -14;
            if (num < 0) {
                System.out.println("Number is Negative");
            }
            else {
                System.out.println("Number is not negative");
            }
        }
}
```

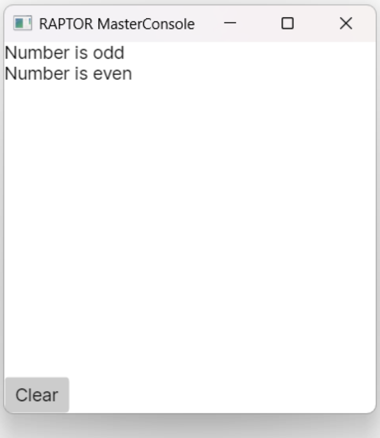
Output:

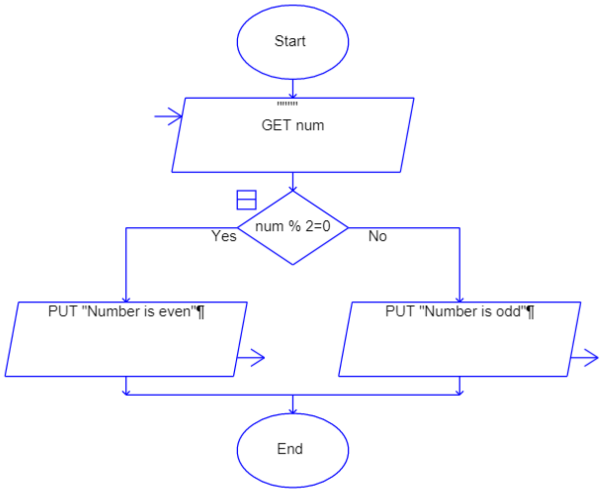
PS F:\Java Code> javac NegativeNumber.java
PS F:\Java Code> java NegativeNumber
Number is Negative

3. Check Odd or Even Number:

- **Task:** Create a flowchart to determine whether a number is odd or even.
- **Next Step:** Write a Java program that checks if a predefined number is odd or even. Use an if-else statement and the modulus operator (%) to determine whether the number is divisible by 2 or not.

Flowchart:





Java Code:

```
public class EvenOdd {
        public static void main(String[] args) {
                int num = 7;
                if (num %2==0) {
                        System.out.println("Number is Even");
                }
                else {
                        System.out.println("Number is Odd");
                }
        }
}
```

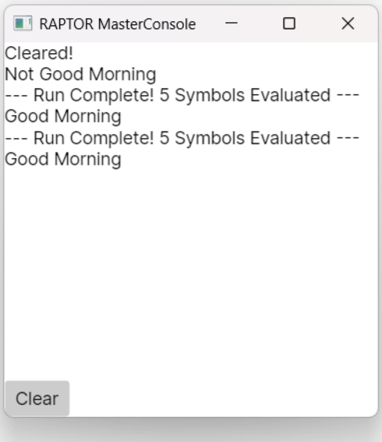
Output:

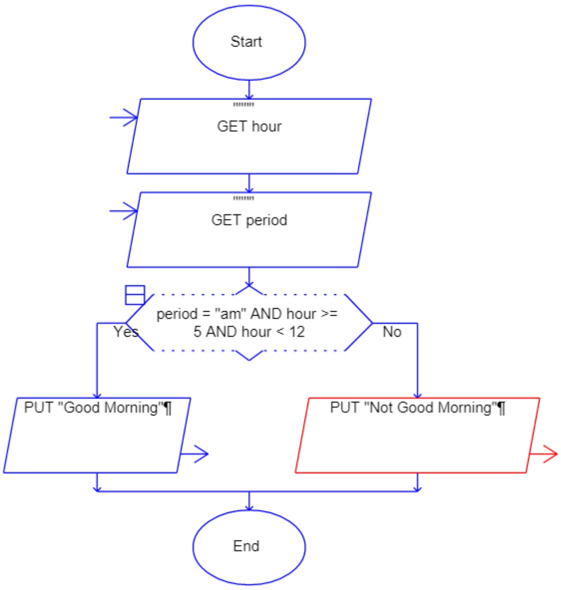
Number is Negative
PS F:\Java Code> javac EvenOdd.java
PS F:\Java Code> java EvenOdd
Number is Odd

4. Display Good Morning Message Based on Time:

- **Task:** Create a flowchart to display a "Good Morning" message based on a given time.
- **Next Step:** Write a Java program that displays a "Good Morning" message if the predefined time is between 5 AM and 12 PM. Use an if statement to implement the logic.

Flowchart:





Java Code:

```
public class GoodMorning {
    public static void main(String[] args) {
        int time = 10;
        if (time >=5 && time <=12) {
            System.out.println("Good Morning");
        }
    }

}
```

Output:

PS F:\Java Code> javac GoodMorning.java

PS F:\Java Code> java GoodMorning

Good Morning

Java Code:

```
public class GoodMorning {
	public static void main(String[] args) {
		int time = 10;
		String period = "am";
		if (period.equals("am") && time >=5 && time <=12) {
			System.out.println("Good Morning");
		}
	}
}
```

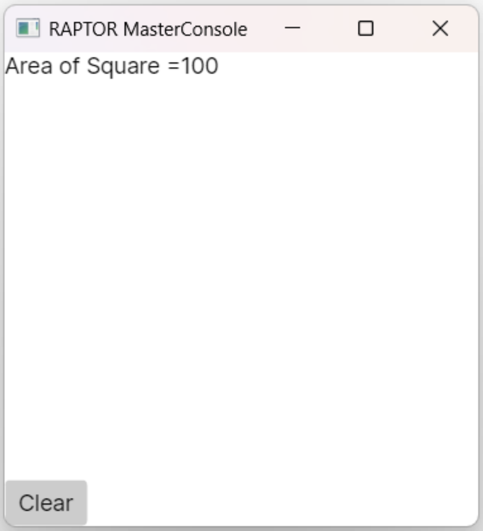
Output:

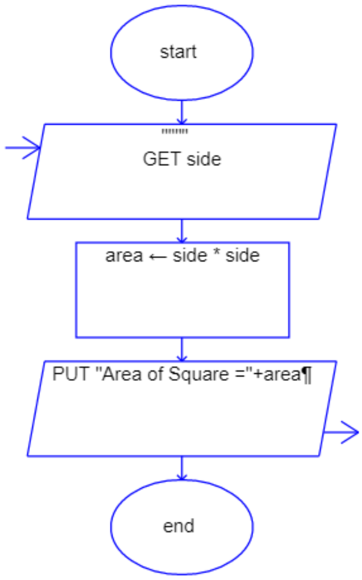
```
PS F:\Java Code> javac GoodMorning.java
PS F:\Java Code> java GoodMorning
Good Morning
```

5. Print Area of a Square:

- **Task:** Create a flowchart to calculate and print the area of a square.
- **Next Step:** Write a Java program that calculates the area of a square using the formula area = side * side. Use a predefined side length.

Flowchart:





Java Code:

```
public class SquareArea {
        public static void main(String[] args) {
            int side = 10;
            int area = side*side;
            System.out.println("Area of Square = " + area );
        }
}
```

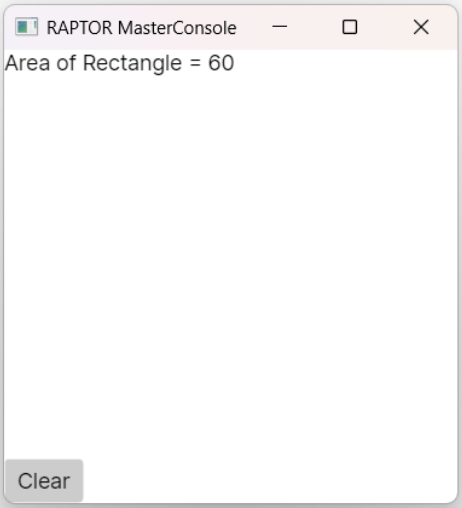
Output:

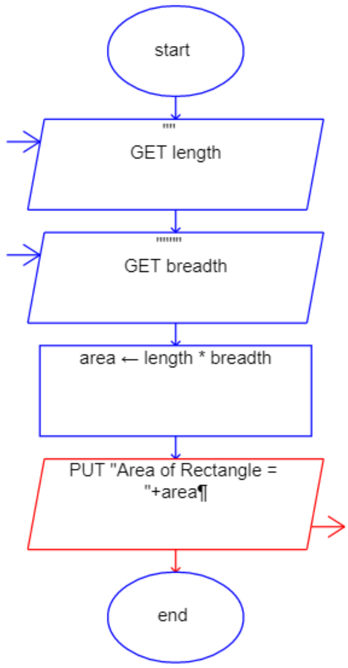
PS F:\Java Code> javac SquareArea.java
PS F:\Java Code> java SquareArea
Area of Square = 100

6. Print Area of a Rectangle:

- **Task:** Create a flowchart to calculate and print the area of a rectangle.
- **Next Step:** Write a Java program that calculates the area of a rectangle using the formula area = length * width. Use predefined values for length and width.

Flowchart:





Java Code:

```
public class RectangleArea {
        public static void main(String[] args) {

            int length = 10;
            int width = 9;
            int area = length*width;

            System.out.println("Area of Rectangle = " + area );
        }
}
```

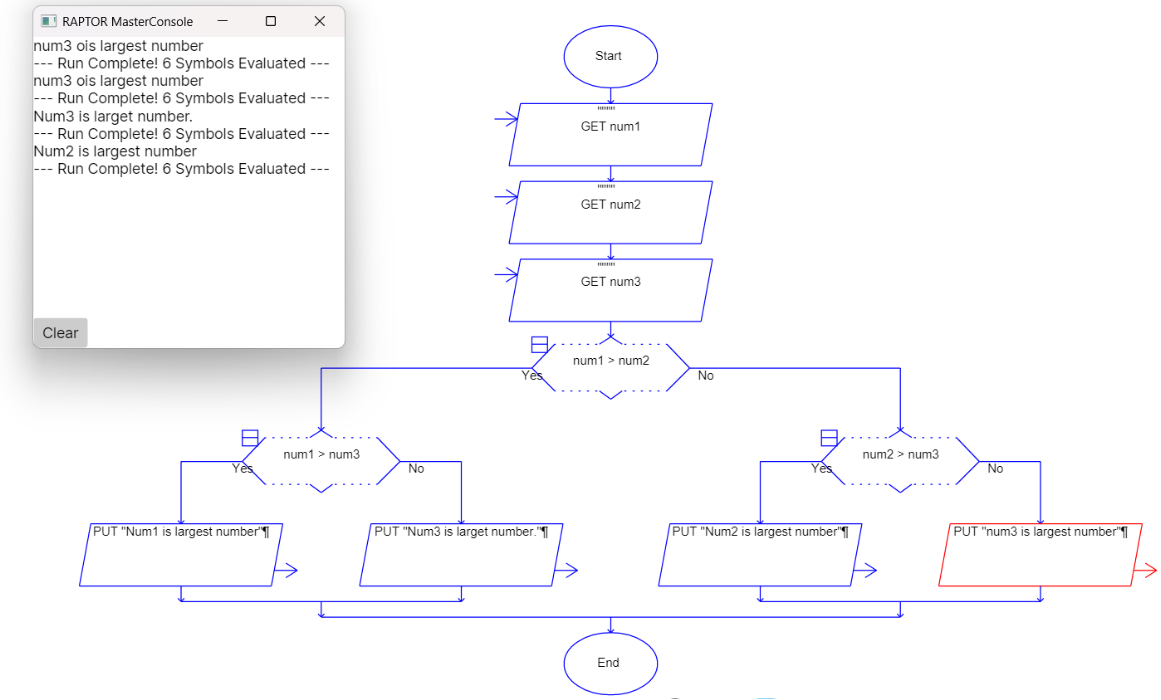
Output:

PS F:\Java Code> javac RectangleArea.java
PS F:\Java Code> java RectangleArea
Area of Rectangle = 90

7. Find the Largest of Three Numbers:

- **Task:** Create a flowchart to find the largest of three numbers.
- **Next Step:** Write a Java program that finds and prints the largest of three predefined numbers using if-else statements.

Flowchart:



Java Code:

```
public class LargestNumber {
                public static void main(String[] args) {
                        int num1 = 10;
                        int num2 = 9;
                        int num3 = 17;
                        if (num1>num2 && num1>num3) {
                                System.out.println(num1 + " is largest number");
                        }
                        else if (num2>num1  && num2>num3) {
                                System.out.println(num2 + " is largest number");
                        }
                        else {
                                System.out.println(num3 + " is largest number");
                        }
                }
}
```

Output:

```
PS F:\Java Code> javac LargestNumber.java
PS F:\Java Code> java LargestNumber
17 is largest number
PS F:\Java Code>
```